THE DEGREE-28 POLYNOMIAL OF A GENUS-4 LINEARISED OPERATOR, THE QUADRATIC-FORM TORSOR, AND THE SYMPLECTIC MONODROMY $\mathrm{Sp}_6(\mathbb{F}_2)$

RUQING CHEN

ABSTRACT. Let $q = 2^k$ and let $\mathfrak{q}_\lambda(x) = \mathsf{T}(x) + 2\,\mathrm{Tr}(\lambda_9 x^9 + \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$ be the $\mathbb{Z}/4$-valued quadratic form attached to the supersingular genus-4 curves $y^2 + y = \lambda_9 x^9 + \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ over $\mathbb{F}_q$, T the Witt-vector trace of the Teichmüller representative. Its polar operator T'_λ is a linearised operator with six-dimensional kernel $\mathcal{K}_\lambda$ and the Gauss sums of $\mathfrak{q}_\lambda$ over $\mathbb{F}_{q^{2m}}$ are governed by the coset $\mathcal{C}_\lambda = T'^{-1}_\lambda(1)$ of 64 points. We prove that $L'_\lambda = T'^8_\lambda$ satisfies $L'_\lambda(x) = x^8 B_\lambda(x)^2 + B_\lambda(x) + 1$ with the explicit polynomial $B_\lambda(x) = \lambda_9^4 x^{28} + \lambda_5^4 x^{12} + \lambda_9^2 x^{10} + \lambda_3^4 x^4 + \lambda_5^2 x^2 + \lambda_9 x + 1$, the exact analogue of the sextic of the genus-2 hexad; that

$$s_{2m}(\lambda) = \sum_{x \in \mathbb{F}_{q^{2m}}} i^{\mathfrak{q}_\lambda(x)} = q^m \sum_{w \in \mathcal{C}_\lambda \cap \mathbb{F}_{q^m}} (-1)^{[B_\lambda(w) \neq 0]}\, i^{\mathfrak{q}^{(m)}_\lambda(w)};$$

and that $\mathcal{C}_\lambda$ is canonically the torsor of quadratic forms on $\mathcal{K}_\lambda$ polarising a canonical symplectic form ω_λ, the 28 roots of B_λ being the forms of Arf invariant 1 and the 36 roots of $x^8 B_\lambda + 1$ those of Arf invariant 0. Consequently $|s_{2m}| = 8q^m$ whenever $\mathcal{C}_\lambda \subset \mathbb{F}_{q^m}$. The Frobenius acts on $\mathcal{K}_\lambda$ through $\mathrm{Sp}_6(\mathbb{F}_2)$; by a right-gcd computation in the twisted polynomial ring $\mathbb{F}_q\{\tau\}$ we determine its $\mathrm{GL}_6(\mathbb{F}_2)$-type for all λ with $q \le 512$: all 24 types of $\mathrm{Sp}_6(\mathbb{F}_2)$ occur, with the frequencies predicted by equidistribution, and the counts are not polynomial in q. An octad structure (the $S_8 = O_6^+(2)$ picture) exists only for the 79% of parameters whose Frobenius type occurs in S_8, and is never canonical; B_λ and the torsor exist for every parameter.

1. Introduction

In Parts I–IV of the author's work on prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$ [4, 5, 6, 7] the degree-6 variance was reduced to the $\mathbb{Z}/4$-valued quadratic forms

$$\mathfrak{q}_\lambda(x) = \mathsf{T}(x) + 2\,\mathrm{Tr}(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$$

on the fields $\mathbb{F}_{q^{2m}}$, where $\mathsf{T}(x) = \mathrm{Tr}_{W_2(\mathbb{F}_{2^n})/W_2(\mathbb{F}_2)}[x]$ is the Witt-vector trace of the Teichmüller representative. These are the forms whose Gauss sums are the traces of Frobenius on the rank-four Witt–Artin–Schreier sheaves of Brylinski and Katz [3, 10], and whose signs are the Brown invariants of [2]. Part III showed that the radical of the polar form of $\mathfrak{q}_\lambda$ is the four-dimensional space $\mathcal{K}_\lambda$ of pairwise sums of the six roots of the sextic $B_\lambda(u) = \lambda_5^2 u^6 + \lambda_3^2 u^2 + \lambda_5 u + 1$, through the identity $L'_\lambda = x^4 B_\lambda^2 + B_\lambda + 1$ for the linearised polynomial L'_λ of the polar operator, so that the Frobenius acts on $\mathcal{K}_\lambda$ through $S_6 \cong \mathrm{Sp}_4(\mathbb{F}_2)$ (the hexad); Part IV proved the *hexad formula* expressing the Gauss sum over $\mathbb{F}_{q^{2m}}$ as a signed sum of at most sixteen fourth roots of unity indexed by the coset $T'^{-1}_\lambda(1)$, with sign $+1$ at the six roots and -1 at the ten synthematic totals.

The present paper treats the next member of the Gold family, the supersingular genus-4 curves

$$y^2 + y = \lambda_9 x^9 + \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x, \qquad \lambda_9 \neq 0, \tag{1}$$

Date: October 11, 2026.

Working draft. Code and data: directory `octad_lab` of the author's Legendre-interval project; the scripts named in Section 9 reproduce every number in this paper.

of van der Geer and van der Vlugt [9], whose quadratic form $\mathfrak{q}_\lambda$ has a polar operator with six-dimensional kernel. The question we started from was whether the hexad of genus 2 becomes an *octad* in genus 4: the 63 nonzero vectors of $\mathbb{F}_2^6$ are the 28 duads and 35 tetrads of an 8-set, giving $S_8 \cong O_6^+(2) \subset \mathrm{Sp}_6(\mathbb{F}_2)$. The answer is that the octad is the wrong structure and that the right one is simpler and more uniform. Our results are the following.

(1) The polynomial $B_\lambda(x) = \lambda_9^4 x^{28} + \lambda_5^4 x^{12} + \lambda_9^2 x^{10} + \lambda_3^4 x^4 + \lambda_5^2 x^2 + \lambda_9 x + 1$ satisfies $L'_\lambda = x^8 B_\lambda^2 + B_\lambda + 1$ (Theorem 3.1), so the 64-point coset $\mathcal{C}_\lambda = T'^{-1}_\lambda(1)$ splits canonically into the 28 roots of B_λ and the 36 roots of $x^8 B_\lambda + 1$.

(2) The coset-sum theorem (Theorem 4.1): $s_{2m}(\lambda) = q^m \sum_{w \in \mathcal{C}_\lambda \cap \mathbb{F}_{q^m}} \varepsilon(w) i^{\mathfrak{q}_\lambda^{(m)}(w)}$ with $\varepsilon(w) = +1$ iff $B_\lambda(w) = 0$. The proof is that of Part IV with two more Newton power sums and one more squaring: the key identities are $\gamma_w^4 = B_\lambda(w)$ and $(w\gamma_w)^2 = \gamma_w$.

(3) The torsor theorem (Theorem 5.1): $\mathcal{K}_\lambda$ carries a canonical nondegenerate alternating form $\omega_\lambda(a,b) = \mathrm{Tr}(d_b T'_\lambda(d_a))$, and $w \mapsto Q_w$, $Q_w(k) = [\varepsilon(w+k) \neq \varepsilon(w)]$, identifies $\mathcal{C}_\lambda$ with the torsor of quadratic forms on $\mathcal{K}_\lambda$ with polar form ω_λ; $\mathrm{Arf}\, Q_w = 1$ iff $B_\lambda(w) = 0$. In genus 2 the same statements hold with $6 = 2^3 - 2$ and $10 = 2^3 + 2$: the hexad roots are the forms of Arf invariant 1. Hence $s_{2m} = \pm 8 q^m i^p$ whenever $\mathcal{C}_\lambda \subset \mathbb{F}_{q^m}$ (Corollary 5.2).

(4) The Frobenius $x \mapsto x^q$ acts on $\mathcal{K}_\lambda$ as an element of $\mathrm{Sp}(\mathcal{K}_\lambda, \omega_\lambda) \cong \mathrm{Sp}_6(\mathbb{F}_2)$. Its conjugacy class in $\mathrm{GL}_6(\mathbb{F}_2)$ (its *type*) is computed for every λ with $q \leq 512$ by right gcds in the twisted polynomial ring $\mathbb{F}_q\{\tau\}$, without field extensions (Section 6). The 24 types of $\mathrm{Sp}_6(\mathbb{F}_2)$ all occur, with the frequencies predicted by equidistribution in $\mathrm{Sp}_6(\mathbb{F}_2)$ to three decimals at $q = 512$ (Table 1); unlike the eleven class counts of genus 2, the counts are not polynomials in q on a parity class of k.

(5) Octads (Section 7): a Frobenius-stable octad in $\mathcal{C}_\lambda$ exists only if the Frobenius type occurs in S_8, which happens for a proportion of parameters tending to $1146880/1451520 = 0.7901$; for $y^2 + y = x^9 + \lambda_1 x$ over $\mathbb{F}_{2^k}$, $9 \nmid k$, the Frobenius has order 9 on $\mathcal{K}_\lambda$ and no octad exists. When octads exist they are not unique (between 3 and 7 for the parameters over $\mathbb{F}_4$, $\mathbb{F}_8$ that have any), and no octic with polynomial coefficients divides $L'_\lambda + 1$. The 28 roots of B_λ are the duads of every octad through a plus-type point.

Everything stated as a theorem is proved below; the statements about Frobenius types for $q \leq 512$ and the frequencies are computations.

2. The form, its polar operator, and the coset

Let $q = 2^k$, $\lambda = (\lambda_9, \lambda_5, \lambda_3, \lambda_1) \in \mathbb{F}_q^4$, $\lambda_9 \neq 0$, $f_\lambda(x) = \lambda_9 x^9 + \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$. For a finite field $E \supset \mathbb{F}_q$ of degree n over $\mathbb{F}_2$ put

$$\mathfrak{q}_\lambda^E(x) = \mathsf{T}_E(x) + 2\,\mathrm{Tr}_{E/\mathbb{F}_2}(f_\lambda(x)) \in \mathbb{Z}/4, \qquad \mathsf{T}_E(x) = \mathrm{Tr}_{W_2(E)/W_2(\mathbb{F}_2)}([x]),$$

and $s_E(\lambda) = \sum_{x \in E} i^{\mathfrak{q}_\lambda^E(x)}$; we write $\mathfrak{q}_\lambda^{(m)}$ and $s_m(\lambda)$ when $E = \mathbb{F}_{q^m}$. Since the first Witt component of $\mathrm{Tr}_{W_2(E)}[x]$ is $\mathrm{Tr}_{E/\mathbb{F}_2}(x)$, we have $\mathfrak{q}_\lambda^E(x) \equiv \mathrm{Tr}_{E/\mathbb{F}_2}(x) \pmod 2$. Define

$$T'_\lambda(x) = \lambda_9 x^8 + \lambda_5 x^4 + \lambda_3 x^2 + x + (\lambda_3 x)^{1/2} + (\lambda_5 x)^{1/4} + (\lambda_9 x)^{1/8}, \tag{2}$$

$$L'_\lambda(x) = T'_\lambda(x)^8 = \lambda_9^8 x^{64} + \lambda_5^8 x^{32} + \lambda_3^8 x^{16} + x^8 + \lambda_3^4 x^4 + \lambda_5^2 x^2 + \lambda_9 x. \tag{3}$$

Lemma 2.1 (polar identity)**.** *For $x, y \in E$: $\mathfrak{q}_\lambda^E(x+y) = \mathfrak{q}_\lambda^E(x) + \mathfrak{q}_\lambda^E(y) + 2\,\mathrm{Tr}_{E/\mathbb{F}_2}\big(y\,T'_\lambda(x)\big)$.*

Proof. In W_2 of a ring of characteristic 2, $[x] + [y] = (x + y, xy) = [x + y] + V[xy]$ and $\mathrm{Tr}_{W_2(E)/W_2(\mathbb{F}_2)} V[a] = 2\,\mathrm{Tr}(a)$; so $\mathsf{T}(x+y) = \mathsf{T}(x) + \mathsf{T}(y) + 2\,\mathrm{Tr}(xy)$. For a Gold monomial,

$(x+y)^{2^i+1}+x^{2^i+1}+y^{2^i+1}=x^{2^i}y+xy^{2^i}$ and $\mathrm{Tr}(\lambda xy^{2^i})=\mathrm{Tr}((\lambda x)^{2^{-i}}y)$. Adding the three monomials of f_λ to $2\,\mathrm{Tr}(xy)$ gives $2\,\mathrm{Tr}(yT'_\lambda(x))$. □

Thus T'_λ is $\mathbb{F}_2$-linear, commutes with the Frobenius $x\mapsto x^q$ (its coefficients lie in $\mathbb{F}_q$), and

$$\mathcal{K}_\lambda=\ker T'_\lambda=\ker L'_\lambda,\qquad \mathcal{C}_\lambda=T'^{-1}_\lambda(1)=\{x:\ L'_\lambda(x)=1\}$$

are, over $\overline{\mathbb{F}}_2$, an $\mathbb{F}_2$-space of dimension 6 (as L'_λ is a separable 2-linearised polynomial of 2-degree 6: its derivative is $\lambda_9\neq 0$) and an affine space under it with 64 points. $\mathcal{K}_\lambda$ is the radical of the polar form of $\mathfrak{q}^E_\lambda$ on every E, and the Frobenius acts on $\mathcal{K}_\lambda$ and on $\mathcal{C}_\lambda$.

Lemma 2.2. *$\mathfrak{q}^E_\lambda$ restricted to $\mathcal{K}_\lambda\cap E$ is additive with values in $\{0,2\}$; hence $\mathrm{Tr}_{E/\mathbb{F}_2}(k)=0$ for all $k\in\mathcal{K}_\lambda\cap E$, and all points of $\mathcal{C}_\lambda\cap E$ have the same absolute trace.*

Proof. By Lemma 2.1 the polar form vanishes on $\mathcal{K}_\lambda$, so $\mathfrak{q}^E_\lambda|_{\mathcal{K}_\lambda\cap E}$ is a homomorphism from an elementary abelian 2-group to $\mathbb{Z}/4$, hence 2-torsion valued. As $\mathfrak{q}\equiv\mathrm{Tr}\pmod 2$, $\mathrm{Tr}(k)=0$. Differences of points of $\mathcal{C}_\lambda$ lie in $\mathcal{K}_\lambda$. □

3. The polynomial of degree 28

Theorem 3.1. *Let $B_\lambda(x)=\lambda_9^4x^{28}+\lambda_5^4x^{12}+\lambda_9^2x^{10}+\lambda_3^4x^4+\lambda_5^2x^2+\lambda_9x+1\in\mathbb{F}_q[x]$. Then*

$$L'_\lambda(x)=x^8B_\lambda(x)^2+B_\lambda(x)+1,\qquad i.e.\qquad L'_\lambda+1=B_\lambda\cdot(x^8B_\lambda+1). \tag{4}$$

Proof. $x^8B_\lambda^2=\lambda_9^8x^{64}+\lambda_5^8x^{32}+\lambda_9^4x^{28}+\lambda_3^8x^{16}+\lambda_5^4x^{12}+\lambda_9^2x^{10}+x^8$; adding B_λ cancels the terms x^{28},x^{12},x^{10} and leaves $L'_\lambda+1$. □

Corollary 3.2. *The two factors in (4) are coprime and $L'_\lambda+1$ is separable, so $\mathcal{C}_\lambda$ is the disjoint union of the 28 roots of B_λ and the 36 roots of $x^8B_\lambda+1$, all simple. Both sets are stable under the Frobenius.*

Remark 3.3. $Y=x^8B_\lambda$ is a polynomial solution of $Y^2+Y=x^8(L'_\lambda+1)$; the only other solution is $Y+1$, not divisible by x^8. So B_λ is determined by L'_λ, and is produced by the coefficient recursion $b_d=c_d+b^2_{(d-8)/2}$ (c_d the coefficients of $L'_\lambda+1$), which terminates because of the shape (3). The same recursion with x^4 in place of x^8 gives the sextic of Part III from the genus-2 operator. For the Gold family with exponents 2^i+1, $i\le n$, one obtains in this way $L'=x^{2^n}B^2+B+1$ with $\deg B=2^{2n-1}-2^{n-1}$, the number of quadratic forms of Arf invariant 1 on $\mathbb{F}_2^{2n}$ with a given polar form; Theorem 5.1 explains why.

4. The coset-sum theorem

Theorem 4.1. *Let $m\ge 1$ and put $\varepsilon(w)=+1$ if $B_\lambda(w)=0$ and $\varepsilon(w)=-1$ otherwise. Then*

$$s_{2m}(\lambda)=q^m\sum_{w\in\mathcal{C}_\lambda\cap\mathbb{F}_{q^m}}\varepsilon(w)\,i^{\mathfrak{q}^{(m)}_\lambda(w)}. \tag{5}$$

Moreover, for any $c\in\mathbb{F}_{q^{2m}}$ with $c+c^{q^m}=w\in\mathcal{C}_\lambda\cap\mathbb{F}_{q^m}$ one has $\mathfrak{q}^{(2m)}_\lambda(c)=\mathfrak{q}^{(m)}_\lambda(w)+2[\varepsilon(w)=-1]$, and the sum of $i^{\mathfrak{q}^{(2m)}_\lambda}$ over a coset $c+\mathbb{F}_{q^m}$ with $c+c^{q^m}\notin\mathcal{C}_\lambda$ vanishes.

Proof. Write $V=\mathbb{F}_{q^{2m}}$, $U=\mathbb{F}_{q^m}$, $\mathfrak{q}=\mathfrak{q}^{(2m)}_\lambda$, $\mathrm{Tr}=\mathrm{Tr}_{V/\mathbb{F}_2}$, $\varphi(x)=x^{q^m}$, so $\mathrm{Tr}_{V/U}=\varphi+1$.

Step 1 ($\mathfrak{q}$ on U). For $u\in U$, $\mathsf{T}_V(u)=\mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}([u]+[u])=\mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}(0,u^2)=2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u)$, and $2\,\mathrm{Tr}(f_\lambda(u))=4\,\mathrm{Tr}_{U/\mathbb{F}_2}(f_\lambda(u))=0$. So $\mathfrak{q}(u)=2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u)$.

Step 2 (cosets of U). For $c\in V$, $u\in U$, Lemma 2.1 and Step 1 give $\mathfrak{q}(c+u)=\mathfrak{q}(c)+2\,\mathrm{Tr}_{U/\mathbb{F}_2}\big(u\,(1+(\varphi+1)T'_\lambda(c))\big)$, hence $\sum_{u\in U}i^{\mathfrak{q}(c+u)}=q^m\,i^{\mathfrak{q}(c)}\,[(\varphi+1)T'_\lambda(c)=1]$ and $\mathfrak{q}$ is constant on the cosets that contribute. Since T'_λ commutes with φ, $(\varphi+1)T'_\lambda(c)=$

$T'_\lambda(c+c^{q^m})$, and $\varphi+1$ maps V onto U with kernel U; so the contributing cosets correspond bijectively to $w = c + c^{q^m} \in \mathcal{C}_\lambda \cap U$ and $s_{2m} = q^m \sum_w i^{\mathfrak{q}(c_w)}$. It remains to prove $\mathfrak{q}(c) = \mathfrak{q}^{(m)}_\lambda(w) + 2[B_\lambda(w) \neq 0]$.

Step 3 (the Witt value of c). Let $N = c\,c^{q^m} \in U$. By transitivity of the Witt trace and $\varphi[c] = [c^{q^m}]$, $\mathsf{T}_V(c) = \mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}([c]+[c^{q^m}]) = \mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}(w, N) = \mathsf{T}_U(w) + 2\,\mathrm{Tr}_{U/\mathbb{F}_2}(N)$. The power sums $p_j = c^j + c^{jq^m}$ satisfy $p_j = wp_{j-1} + Np_{j-2}$, $p_0 = 0$, $p_1 = w$, whence $p_3 = w^3+Nw$, $p_5 = w^5+Nw^3+N^2w$, $p_9 = w^9+Nw^7+N^2w^5+N^4w$. Using $\mathrm{Tr}_{U/\mathbb{F}_2}(N^{2^j}a) = \mathrm{Tr}_{U/\mathbb{F}_2}(Na^{2^{-j}})$,

$$\mathfrak{q}(c) = \mathfrak{q}^{(m)}_\lambda(w) + 2\,\mathrm{Tr}_{U/\mathbb{F}_2}(N\gamma_w),$$
$$\gamma_w = 1 + \lambda_9 w^7 + (\lambda_9 w^5)^{1/2} + (\lambda_9 w)^{1/4} + \lambda_5 w^3 + (\lambda_5 w)^{1/2} + \lambda_3 w.$$

Squaring twice,

$$\gamma_w^2 = 1 + \lambda_9^2 w^{14} + \lambda_9 w^5 + (\lambda_9 w)^{1/2} + \lambda_5^2 w^6 + \lambda_5 w + \lambda_3^2 w^2, \qquad \gamma_w^4 = B_\lambda(w).$$

From $T'_\lambda(w) = 1$, i.e. $\lambda_9 w^8 + \lambda_5 w^4 + \lambda_3 w^2 + w = 1 + (\lambda_3 w)^{1/2} + (\lambda_5 w)^{1/4} + (\lambda_9 w)^{1/8}$,

$$w\gamma_w = 1 + (\lambda_3 w)^{1/2} + (\lambda_5 w)^{1/4} + (\lambda_9 w)^{1/8} + (\lambda_9 w^7)^{1/2} + (\lambda_9 w^5)^{1/4} + (\lambda_5 w^3)^{1/2},$$

and $(w\gamma_w)^4 = 1 + \lambda_3^2 w^2 + \lambda_5 w + (\lambda_9 w)^{1/2} + \lambda_9^2 w^{14} + \lambda_9 w^5 + \lambda_5^2 w^6 = \gamma_w^2$. Squaring being injective, $(w\gamma_w)^2 = \gamma_w$, so $\gamma_w \in \{0, w^{-2}\}$ ($w \neq 0$ as $T'_\lambda(0) = 0$), and $\gamma_w = 0$ iff $B_\lambda(w) = 0$.

If $B_\lambda(w) = 0$ then $\mathfrak{q}(c) = \mathfrak{q}^{(m)}_\lambda(w)$. If $B_\lambda(w) \neq 0$ then $N\gamma_w = c\,c^{q^m}/w^2 = v\,v^{q^m}$ with $v = c/w$; as $v + v^{q^m} = 1$, $N\gamma_w = v^2 + v$ with $v \notin U$, so the Artin–Schreier polynomial $Y^2 + Y + (v^2 + v)$ has no root in U, $\mathrm{Tr}_{U/\mathbb{F}_2}(v^2 + v) = 1$, and $\mathfrak{q}(c) = \mathfrak{q}^{(m)}_\lambda(w) + 2$. □

Remark 4.2. Nothing in the proof refers to roots of B_λ as a configuration; only the Gold shape of f_λ (bilinearity of the polar form) and the identities $\gamma_w^4 = B_\lambda(w)$, $(w\gamma_w)^2 = \gamma_w$ are used. In Part IV the corresponding identities are $\gamma_w^2 = B_\lambda(w)$, $(w\gamma_w)^2 = \gamma_w$. The sign $\varepsilon(w)$ does not depend on λ_1, nor on the field $\mathbb{F}_{q^m}$ over which w is considered.

5. The torsor of quadratic forms

Fix m with $\mathcal{C}_\lambda \subset \mathbb{F}_{q^m}$ (any multiple of the order of the Frobenius on $\mathcal{K}_\lambda$), $V = \mathbb{F}_{q^{2m}}$, $U = \mathbb{F}_{q^m}$. For $a \in \mathcal{K}_\lambda$ let $d_a \in V$ be any *lift*, $d_a + d_a^{q^m} = a$; for $w \in \mathcal{C}_\lambda$ let c_w be a lift of w.

Theorem 5.1. (i) $\omega_\lambda(a, b) = \mathrm{Tr}_{V/\mathbb{F}_2}\big(d_b\,T'_\lambda(d_a)\big)$ *is a well-defined nondegenerate alternating form on $\mathcal{K}_\lambda$, independent of m.*

(ii) *For $w \in \mathcal{C}_\lambda$, $Q_w(k) = [\varepsilon(w + k) \neq \varepsilon(w)]$ is a quadratic form on $\mathcal{K}_\lambda$ with polar form ω_λ; explicitly*

$$Q_w(k) = \tfrac{1}{2}\big[\mathfrak{q}^{(2m)}_\lambda(d_k) - \mathfrak{q}^{(m)}_\lambda(k)\big] + \mathrm{Tr}_{V/\mathbb{F}_2}\big(d_k\,T'_\lambda(c_w)\big) \pmod 2. \tag{6}$$

(iii) *$Q_{w+k} = Q_w + \omega_\lambda(k, \cdot)$, and $w \mapsto Q_w$ is a bijection from $\mathcal{C}_\lambda$ onto the set of quadratic forms on $\mathcal{K}_\lambda$ with polar form ω_λ.*

(iv) $\mathrm{Arf}(Q_w) = 1$ (28 *zeros*) *iff* $\varepsilon(w) = +1$ *iff* $B_\lambda(w) = 0$; $\mathrm{Arf}(Q_w) = 0$ (36 *zeros*) *iff* $x^8 B_\lambda(w) + 1 = 0$.

Proof. (ii) and well-definedness. Let $c = c_w$ and $d = d_k$. By Theorem 4.1 (applied to w with lift c and to $w + k$ with lift $c + d$) and Lemma 2.1,

$$\varepsilon(w+k)\varepsilon(w) = i^{\mathfrak{q}(c+d)-\mathfrak{q}(c)-\mathfrak{q}^{(m)}(w+k)+\mathfrak{q}^{(m)}(w)} = i^{\mathfrak{q}(d)+2\,\mathrm{Tr}(dT'(c))-\mathfrak{q}^{(m)}(k)-2\,\mathrm{Tr}_{U/\mathbb{F}_2}(kT'(w))},$$

with $T'(w) = 1$ and $\mathrm{Tr}_{U/\mathbb{F}_2}(k) = 0$ (Lemma 2.2). All exponents are even ($\mathfrak{q}(d) \equiv \mathrm{Tr}_{V/\mathbb{F}_2}(d) = \mathrm{Tr}_{U/\mathbb{F}_2}(k) = 0$, and $\mathfrak{q}^{(m)}(k) \in \{0, 2\}$), which gives (6). Replacing d by $d + u$, $u \in U$, changes $\mathfrak{q}(d)$ by $\mathfrak{q}(u) + 2\,\mathrm{Tr}(uT'(d)) = 2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u) + 2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u\,T'(k)) = 2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u)$ and $\mathrm{Tr}(dT'(c))$ by $\mathrm{Tr}_{U/\mathbb{F}_2}(u\,T'(w)) = \mathrm{Tr}_{U/\mathbb{F}_2}(u)$; the changes cancel. So the

right side of (6) is a function of k; as a function of d it is the sum of the quadratic form $\frac{1}{2}[\mathfrak{q}(d)-\mathfrak{q}^{(m)}(d+d^{q^m})]$ (note $\mathfrak{q}^{(m)}|_{\mathcal{K}_\lambda}$ is additive), whose polar form is $\mathrm{Tr}(d'T'(d))$, and of a linear form. Hence Q_w is a quadratic form on $\mathcal{K}_\lambda \cong \widetilde{V}/U$, $\widetilde{V}=\{d\in V : d+d^{q^m}\in\mathcal{K}_\lambda\}$, with polar form ω_λ, which is thereby well defined.

(i) Alternating: $2\,\mathrm{Tr}(dT'(d)) = \mathfrak{q}(2d)-2\mathfrak{q}(d) \equiv 2\mathfrak{q}(d)$, so $\omega_\lambda(a,a)\equiv\mathfrak{q}(d_a)\equiv \mathrm{Tr}_{U/\mathbb{F}_2}(a)=0$. Nondegenerate: the symmetric bilinear form $\beta(x,y)=\mathrm{Tr}(yT'(x))$ on V has radical $\mathcal{K}_\lambda$ (the trace form is nondegenerate). $\widetilde{V}$ has dimension $mk+6$ and contains U and $\mathcal{K}_\lambda$; $U\subset\widetilde{V}^\perp$ because $\beta(u,d)=\mathrm{Tr}_{U/\mathbb{F}_2}(u\,T'(d+d^{q^m}))=0$; and $\dim\widetilde{V}^\perp=\dim V-\dim\widetilde{V}+\dim(\widetilde{V}\cap\mathcal{K}_\lambda)=2mk-(mk+6)+6=mk=\dim U$. So $\widetilde{V}^\perp=U$, $\mathrm{rad}(\beta|_{\widetilde{V}})=U$, and β descends to the nondegenerate form ω_λ on $\widetilde{V}/U\cong\mathcal{K}_\lambda$. Independence of m: ε is independent of m by Theorem 4.1, hence so are Q_w and its polar form.

(iii) $Q_{w+k}(k')=[\varepsilon(w+k+k')\neq\varepsilon(w+k)]=Q_w(k+k')+Q_w(k)=Q_w(k')+\omega_\lambda(k,k')$. The quadratic forms with polar form ω_λ form a torsor under $\mathcal{K}_\lambda^*\cong\mathcal{K}_\lambda$ (via ω_λ), and $w\mapsto Q_w$ is an equivariant map of $\mathcal{K}_\lambda$-torsors, hence bijective.

(iv) $\sum_{k\in\mathcal{K}_\lambda}\varepsilon(w+k)=\varepsilon(w)\sum_k(-1)^{Q_w(k)}=\varepsilon(w)(-1)^{\mathrm{Arf}\,Q_w}2^3$, the Gauss sum of a nondegenerate quadratic form on $\mathbb{F}_2^6$. The left side is $\sum_{w'\in\mathcal{C}_\lambda}\varepsilon(w')=28-36=-8$ by Corollary 3.2. Hence $\varepsilon(w)(-1)^{\mathrm{Arf}\,Q_w}=-1$. □

Corollary 5.2 (sixty-four-point lemma)**.** *If $\mathcal{C}_\lambda\subset\mathbb{F}_{q^m}$ then, for every $\lambda_1\in\mathbb{F}_q$, $s_{2m}(\lambda)=\pm8\,q^m\,i^p$, where $p=\mathfrak{q}_\lambda^{(m)}(w_0)$ for any $w_0\in\mathcal{C}_\lambda$ ($\mathfrak{q}_\lambda^{(m)}$ is constant modulo 2 on $\mathcal{C}_\lambda$).*

Proof. By Lemmas 2.1 and 2.2, $\mathfrak{q}_\lambda^{(m)}(w_0+k)=\mathfrak{q}_\lambda^{(m)}(w_0)+2\ell(k)$ with $\ell=\frac{1}{2}\mathfrak{q}_\lambda^{(m)}|_{\mathcal{K}_\lambda}$ linear. Then $\sum_{\mathcal{C}_\lambda}\varepsilon(w)i^{\mathfrak{q}^{(m)}(w)}=\varepsilon(w_0)i^{\mathfrak{q}^{(m)}(w_0)}\sum_k(-1)^{(Q_{w_0}+\ell)(k)}=\pm8\,i^{\mathfrak{q}^{(m)}(w_0)}$. □

For $q=2$ one finds $s_{14}(1,1,1,\lambda_1)=\mp1024\,i$ and $s_{18}(1,0,0,\lambda_1)=\pm4096\,i$ ($\lambda_1=0,1$), in accordance with $8\cdot2^7$ and $8\cdot2^9$.

Remark 5.3 (genus 2)**.** With the quartic operator of Part III the same proof gives: ω_λ canonical on the four-dimensional $\mathcal{K}_\lambda$, $\mathcal{C}_\lambda$ the torsor of quadratic forms, the six roots of the sextic B_λ the forms of Arf invariant 1 ($6=2^3-2$ zeros), the ten synthematic totals the forms of Arf invariant 0 ($10=2^3+2$). The hexad formula of Part IV is thus the statement $\varepsilon(w)=-(-1)^{\mathrm{Arf}\,Q_w}$, and the sixteen-point lemma of Part IV is Corollary 5.2 with ±4.

6. The Frobenius on $\mathcal{K}_\lambda$

By Theorem 5.1 the Frobenius $F: x\mapsto x^q$ acts on $\mathcal{K}_\lambda$ as an element of $\mathrm{Sp}(\mathcal{K}_\lambda,\omega_\lambda)\cong\mathrm{Sp}_6(\mathbb{F}_2)$. Its *type* is its conjugacy class in $\mathrm{GL}_6(\mathbb{F}_2)$, i.e. the elementary divisors of the $\mathbb{F}_2[z]$-module $(\mathcal{K}_\lambda,F)$; $\mathrm{Sp}_6(\mathbb{F}_2)$ has 30 conjugacy classes and 24 types.

Proposition 6.1. *Let $\mathbb{F}_q\{\tau\}$ be the twisted polynomial ring, $\tau c=c^2\tau$, and $P_\lambda=\lambda_9^8\tau^6+\lambda_5^8\tau^5+\lambda_3^8\tau^4+\tau^3+\lambda_3^4\tau^2+\lambda_5^2\tau+\lambda_9$ (so that $P_\lambda(x)=L'_\lambda(x)$). For $g\in\mathbb{F}_2[z]$,*

$$\dim_{\mathbb{F}_2}\ker\big(g(F)|_{\mathcal{K}_\lambda}\big)=\deg_\tau\mathrm{rgcd}\big(P_\lambda,\ g(\tau^k)\big).$$

Proof. For separable additive polynomials A,B over $\mathbb{F}_q$, $\ker A\subseteq\ker B$ iff $B=RA$ in $\mathbb{F}_q\{\tau\}$ (right divisibility), and the right ideal $A\mathbb{F}_q\{\tau\}+B\mathbb{F}_q\{\tau\}$ is principal, generated by the right gcd D, with $\ker D=\ker A\cap\ker B$ [13, 12]. Since $F=\tau^k$ and $g\in\mathbb{F}_2[z]$ commutes with τ, $g(F)$ is the additive polynomial $g(\tau^k)$, and $\ker g(F)|_{\mathcal{K}_\lambda}=\ker P_\lambda\cap\ker g(\tau^k)$. □

The ranks of $p(F)^j$ for the irreducible $p\in\mathbb{F}_2[z]$ of degree ≤6 and $j\le6$ determine the type; they are computed by right Euclid in $\mathbb{F}_q\{\tau\}$ with no field extension (script

`ore_algebra_k6.py`, checked against companion matrices). The types of all elements of $\mathrm{Sp}_6(\mathbb{F}_2)$, built as the group generated by S_8 and one transvection on the 63-point space, were computed separately (`sp6_module_types.py`): there are 24, of which 20 occur in S_8 and 22 in $O_6^-(2)$; every element of $\mathrm{Sp}_6(\mathbb{F}_2)$ preserves at least one quadratic form with polar form ω.

TABLE 1. The 24 types of $\mathrm{Sp}_6(\mathbb{F}_2)$ ($\Phi_3 = z^2+z+1$, $\Phi_5 = z^4+z^3+z^2+z+1$, $\Phi_7 = (z^3+z+1)(z^3+z^2+1)$, $\Phi_9 = z^6+z^3+1$), the S_8 classes with that type, the proportion of elements of $\mathrm{Sp}_6(\mathbb{F}_2)$ of that type, and the observed proportion of parameters $(\lambda_5, \lambda_3) \in \mathbb{F}_{512}^2$ ($\lambda_9 = 1$) whose Frobenius has that type.

elementary divisors	S_8 class	$\lvert C\rvert/\lvert\mathrm{Sp}_6\rvert$	observed, $q = 512$	octad
Φ_7	$7\,1$	$1/7 = 0.1429$	0.1445	yes
$(z+1)^6$	8	$1/8 = 0.1250$	0.1230	yes
Φ_9	–	$1/9 = 0.1111$	0.1109	no
$(z+1)^2\Phi_5$	$5\,2\,1$	$1/10 = 0.1000$	0.1004	yes
$(z+1)^4\Phi_3$	$4\,3\,1$	$1/12 = 0.0833$	0.0839	yes
$(z+1)^2\Phi_3^2$	$6\,1\,1$	$1/12 = 0.0833$	0.0838	yes
Φ_3^3 (cyclic)	–	$1/12 = 0.0833$	0.0835	no
$\Phi_3\Phi_5$	$5\,3$	$1/15 = 0.0667$	0.0660	yes
$(z+1), (z+1), \Phi_5$	$5\,1^3$	$1/30 = 0.0333$	0.0336	yes
$(z+1)^2, (z+1)^4$	$4\,2\,1\,1$	$1/32 = 0.0313$	0.0309	yes
$(z+1)^2, (z+1)^2, \Phi_3$	$3\,2\,2\,1$	$1/36 = 0.0278$	0.0279	yes
$(z+1), (z+1), \Phi_3^2$	$6\,2$	$1/36 = 0.0278$	0.0281	yes
$(z+1)^2, \Phi_3, \Phi_3$	$3\,3\,2$	$1/36 = 0.0278$	0.0274	yes
Φ_3, Φ_3^2	–	$1/72 = 0.0139$	0.0142	no
$(z+1), (z+1), (z+1)^4$	$4\,2\,2$, $4\,1^4$	$1/96 = 0.0104$	0.0105	yes
$(z+1)^3, (z+1)^3$	$4\,4$	$1/96 = 0.0104$	0.0102	yes
$(z+1), (z+1), \Phi_3, \Phi_3$	$3\,3\,1\,1$	$1/108 = 0.0093$	0.0092	yes
$(z+1), (z+1), (z+1)^2, \Phi_3$	$3\,2\,1^3$	$1/144 = 0.0069$	0.0065	yes
$(z+1)^2, (z+1)^2, (z+1)^2$	$2^3\,1^2$	$1/384 = 0.0026$	0.0027	yes
Φ_3, Φ_3, Φ_3	–	$1/648 = 0.0015$	0.0015	no
$(z+1), (z+1), (z+1)^2, (z+1)^2$	2^4, $2^2\,1^4$	$1/1152 = 0.0009$	0.0008	yes
$(z+1)^{\times 4}, \Phi_3$	$3\,1^5$	$1/2160 = 0.0005$	0.0004	yes
$(z+1)^{\times 4}, (z+1)^2$	$2\,1^6$	$63/\lvert\mathrm{Sp}_6\rvert$	$9/2^{18}$	yes
$(z+1)^{\times 6}$	1^8	$1/\lvert\mathrm{Sp}_6\rvert$	$1/2^{18}$	yes

Proposition 6.2 (computation). *For every $(\lambda_5, \lambda_3) \in \mathbb{F}_q^2$, $\lambda_9 = 1$, $q = 2, 4, \ldots, 512$ (349 524 parameters), the type of F on $\mathcal{K}_\lambda$ is one of the 24 types of $\mathrm{Sp}_6(\mathbb{F}_2)$; at $q = 512$ all 24 occur, with the frequencies of Table 1. The proportion of parameters whose type occurs in S_8 is $0.500, 0.750, 0.750, 0.781, 0.773, 0.791, 0.794, 0.787, 0.790$ for $q = 2, \ldots, 512$, against $1146880/1451520 = 0.7901$ for $\mathrm{Sp}_6(\mathbb{F}_2)$. For no type and no parity of k is the count a polynomial of degree ≤ 2 in q (the quadratic through the three largest q of a parity class misses the smaller ones); the deviations from $\lvert C\rvert q^2/\lvert\mathrm{Sp}_6(\mathbb{F}_2)\rvert$ are of size $O(q)$ (e.g. 37885 against 37449.1 for the type Φ_7 at $q = 512$).*

The agreement with equidistribution in $\mathrm{Sp}_6(\mathbb{F}_2)$ (and the absence of any non-symplectic type, which Theorem 5.1 now explains) is what one expects if the geometric monodromy of the local system $\lambda \mapsto \mathcal{K}_\lambda$ is all of $\mathrm{Sp}_6(\mathbb{F}_2)$ [11]; in genus 2 the eleven counts are exact polynomials (Part III, Table 2), and the failure of polynomiality here is the first qualitative difference between the two cases. Whether a finer invariant than the GL_6-type

(the Sp_6-class, or the Frobenius orbit structure on the $28+36$ points) has polynomial counts is open.

7. Octads

Let E be an 8-set. The even subsets of E modulo complement form $V_8 = \mathbb{F}_2^6$; its 63 nonzero vectors are the 28 duads and 35 tetrads; $\langle x, y\rangle = |x \cap y| \bmod 2$ is a symplectic form and $Q_8(x) = |x|/2 \bmod 2$ a quadratic form of Arf invariant 0 (singular points the tetrads), and $S_8 = O(Q_8) = O_6^+(2)$ has index 36 in $\mathrm{Sp}(V_8)$; the odd subsets modulo complement form the affine space of 8 singletons and 56 triads. (For a 6-set, $Q_6 = |x|/2$ is not well defined and $S_6 = \mathrm{Sp}_4(\mathbb{F}_2)$: genus 2 never sees an orthogonal group.) An *octad* in $\mathcal{C}_\lambda$ is a set of eight points $e_1, \dots, e_8$ with $\sum e_i = 0$ whose duads and tetrads exhaust $\mathcal{K}_\lambda \setminus 0$; it is the same as an identification $\mathcal{K}_\lambda \cong V_8$ together with a base point.

Proposition 7.1. (a) *A Frobenius-stable octad exists in $\mathcal{C}_\lambda$ only if the type of F occurs in S_8. For $\lambda = (1,0,0)$ over $\mathbb{F}_{2^k}$ with $9 \nmid k$, F has order 9 on $\mathcal{K}_\lambda$ and no element of S_8 has order 9: no octad exists. The polynomial $L'_{(1,0,0)} + 1 = x^{64} + x^8 + x + 1$ factors over $\mathbb{F}_2$ into one linear and seven irreducible nonic factors, so no octic with coefficients in $\mathbb{F}_2[\lambda_5, \lambda_3]$ divides $L'_\lambda + 1$.*

(b) *For the parameters over $\mathbb{F}_4$ and $\mathbb{F}_8$ whose type occurs in S_8, the number of Frobenius-stable octads is 7 (type $7\,1$), 6 ($5\,2\,1$), 4 ($4\,3\,1$), 3 ($5\,3$) or 0 (types 8, $6\,1\,1$, $6\,2$, $4\,4$, $4\,2\,1\,1$, $5\,1^3$); their octics have no common shape.*

(c) *If $P \in \mathcal{C}_\lambda$ is a plus-type point, then for every octad through P (identification $\mathcal{K}_\lambda \cong V_8$ with $Q_P = Q_8$) the set $P + \{\text{duads}\}$ is the set of roots of B_λ and $P + \{0\} \cup \{\text{tetrads}\}$ the set of roots of $x^8 B_\lambda + 1$.*

Proof. (a) F permutes a stable octad, so its action on $\mathcal{K}_\lambda \cong V_8$ is that of an element of S_8. For $\lambda = (1,0,0)$, $\mathcal{K}_\lambda \cong \mathbb{F}_2[z]/(\Phi_9)$ as an $\mathbb{F}_2[F]$-module (Φ_9 is the 2-associate of $x^{64}+x^8+x$), so F has order 9 for $9 \nmid k$; the factorisation is a computation. A polynomial family of octics would specialise to a degree-8 factor at $(0,0)$. (b) is a computation (all Frobenius-stable octads are enumerated from the Frobenius orbits on $\mathcal{C}_\lambda$). (c) By Theorem 5.1(iii), $\varepsilon(P+k) = -\varepsilon(P)$ iff $Q_P(k) = 1$ iff k is a duad; $\varepsilon(P) = -1$. □

So the 28 roots of B_λ are the duads of any octad through a plus-type point, which is why they, and not eight points, form the canonical configuration; and a parameter has octads only when its Frobenius type lies in S_8, which Proposition 6.2 says happens for 79% of the parameters, while B_λ and the torsor exist for all of them.

8. Questions

(1) The signs of $s_{2m}(\lambda)$ when $\mathcal{C}_\lambda \cap \mathbb{F}_{q^m}$ is a proper affine subspace: by Theorems 4.1 and 5.1, s_{2m}/q^m is the Gauss sum of $Q_{w_0} + \ell$ restricted to a subspace, hence 0 or $\pm 2^j i^p$; the sign requires the value of Tr on $\mathcal{C}_\lambda$ (the genus-4 trace parity) and the analogue of the hyperplane parameter κ_0 of Part IV, type by type for the 24 types.

(2) The counts of the Frobenius types as functions of q, and the geometric monodromy of $\lambda \mapsto \mathcal{K}_\lambda$.

(3) Which Legendre-interval problem the family (1) governs (degree 10 if the pattern of Parts I–III persists; the degree-8 problem involves x^7, which has 2-weight 3, no quadratic form, and curves $y^2+y = x^7+\dots$ that are not supersingular: their Newton polygons over $\mathbb{F}_2$ have slopes $1/3, 2/3$).

9. Computations

All scripts are in Python (numpy, sympy). `octad_combinatorics.py`: the 63-point geometry, S_8, $\mathrm{Sp}_6(\mathbb{F}_2)$ by closure, orbits, element orders. `genus_check.py`: Newton polygons of $y^2+y=x^7+\dots$, bilinearity tests, module types of S_8, the order-9 obstruction. `ore_algebra_k6.py`, `sp6_module_types.py`, `genus4_class_stats.py`: Proposition 6.1, Table 1, Proposition 6.2. `octad_sum_test.py`: Lemma 2.1 on $\mathbb{F}_{16}$, $\mathbb{F}_{64}$ (all λ) and the fibre sums over $\mathbb{F}_{2^{14}}$, $\mathbb{F}_{2^{18}}$. `find_octic_identity.py`: Proposition 7.1(a),(b). `coset_signs.py`, `b28_identity.py`: ε from one lift per point, the torsor checks and the interpolation that first produced B_λ (all λ for $q\le 8$, samples at $q=16,32$, general λ_9 at $q=4,8$), the identities $\gamma_w^4=B_\lambda(w)$, $(w\gamma_w)^2=\gamma_w$ on every coset point for several parameters, and the exact division (4) in $\mathbb{F}_2[\lambda_9,\lambda_5,\lambda_3][x]$.

References

[1] C. Arf, Untersuchungen über quadratische Formen in Körpern der Charakteristik 2, J. reine angew. Math. 183 (1941), 148–167.

[2] E. H. Brown, Generalizations of the Kervaire invariant, Ann. of Math. 95 (1972), 368–383.

[3] J.-L. Brylinski, Théorie du corps de classes de Kato et revêtements abéliens de surfaces, Ann. Inst. Fourier 33 (1983), 23–38.

[4] R. Chen, Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, I, DOI 10.5281/zenodo.23225896.

[5] R. Chen, Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, II, DOI 10.5281/zenodo.23252762.

[6] R. Chen, Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, III: the hexad, $\mathrm{Sp}_4(\mathbb{F}_2)$ and Witt sums, DOI 10.5281/zenodo.23290724.

[7] R. Chen, Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, IV: point counts on hexad modular curves, Brown invariants of Witt sheaves, and the constant X, DOI 10.5281/zenodo.23299284.

[8] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker, R. A. Wilson, Atlas of Finite Groups, Oxford University Press, 1985.

[9] G. van der Geer, M. van der Vlugt, Reed–Muller codes and supersingular curves. I, Compositio Math. 84 (1992), 333–367.

[10] N. M. Katz, Gauss Sums, Kloosterman Sums, and Monodromy Groups, Ann. of Math. Studies 116, Princeton University Press, 1988.

[11] N. M. Katz, P. Sarnak, Random Matrices, Frobenius Eigenvalues, and Monodromy, AMS Colloquium Publications 45, 1999.

[12] R. Lidl, H. Niederreiter, Finite Fields, 2nd ed., Cambridge University Press, 1997, Chapter 3.4 (linearised polynomials).

[13] O. Ore, On a special class of polynomials, Trans. Amer. Math. Soc. 35 (1933), 559–584.

GUT Geoservice Inc., Montreal, Canada

Email address: `ruqing@hotmail.com`